



Final Review Solutions

Dec 20, 2019

1 Practice Problems

Problem 1.

Let discrete random variables X and Y have the following joint probability mass function: $p_{X,Y}(1,1) = 2/8$, $p_{X,Y}(-1,-1) = 2/8$, $p_{X,Y}(0,0) = 2/8$, $p_{X,Y}(1,-1) = 1/8$, $p_{X,Y}(-1,1) = 1/8$.

1. Find $P(XY > 0)$.

Note that $xy > 0$ if $(x,y) \in \{(1,1), (-1,-1), \}$. So

$$P(XY > 0) = (2/8) + (2/8) = (4/8)$$

2. Find $E[X]$.

$$E[X] = 1(2/8) + (-1)(2/8) + 1(1/8) + (-1)(1/8) = 0.$$

3. Find $E[X|X \geq 0]$.

4. Find $Cov(X,Y)$.

5. Let X_i ($i = 1, 2, \dots, 5$) be independent random variables with the same probability mass function as X . Let $Z = X_1 + X_2 + \dots + X_5$. Find $P(Z = 4)$.

Problem 2.

Let X and Y be jointly distributed random variables whose joint probability density function is given by:

$$f_{X,Y}(x,y) = \begin{cases} x(y+1), & 0 < x < 1, -1 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find $P(XY > 0)$.

Note that $xy > 0$ if $x > 0$ and $y > 0$

$$P(XY > 0) = \int_0^1 \int_0^1 x(y+1) dy dx = \frac{3}{4}.$$

2. Find the marginal probability density function of X , $f_X(x)$.

$$f_X(x) = \int_{-1}^1 x(y+1) dy = 2x \text{ if } 0 < x < 1 \text{ and } f_X(x) = 0 \text{ otherwise}$$

3. Find $E[X]$.

$$E[X] = \int_0^1 x(2x) dx = \frac{2}{3}$$

4. Find $Var[X]$.

5. Find $E[X|X > 1/2]$.

Note that

$$f_{X|X>1/2}(x) = \frac{f_X(x)}{P(X > 1/2)} = \frac{2x}{3/4} = \frac{8x}{3} \text{ if } 1/2 < x < 1 \text{ and } f_X(x) = 0 \text{ otherwise}$$

6. Find the conditional density function $f_{Y|X}(y)$.

7. Are X and Y independent (explain)?

Problem 3.

The download time of large file (for example 1 GB) is an exponential random variable that depends on the congestion level on the local network. Assume that the congestion level on the local network is a continuous uniform random variable Y in the interval $(0,0.9)$ (i.e. $f_Y(y) = 1/0.9$ for $0 < y < 0.9$ and $f_Y(y) = 0$ otherwise). For instance $y = 0.8$ denotes a network congestion level of 80%. Now, assume that the probability density function of the download time (in minutes) given $Y = y$ is given by:

$$f_{T|Y}(t) = (1-y)e^{-(1-y)t} \text{ for } t > 0 \text{ and } f_{T|Y}(t) = 0 \text{ otherwise}$$

1. Assume that $Y = 1/2$. What is the probability that the download time is over 2 minutes ($P(T > 2|Y = 1/2)$)?

$$P(T > 2|Y = 1/2) = \int_2^\infty \frac{e^{-t/2}}{2} dt = e^{-1}.$$

2. What is the probability that the download time is over 2 minutes($P(T > 2)$)?

To find $P(T > 2)$, we condition to find $P(T > 2|Y = y)$ and use the total law of probability.

$$P(T > 2|Y = y) = e^{-(1-y)2} = 0.368.$$

Then:

$$P(T > 2) = \int_0^{0.9} (1/0.9)e^{-(1-y)2} dy = 0.380.$$

3. What is the probability density function of the download time ($f_T(t)$)?

Let's first obtain the cumulative distribution $F_T(t)$ using the above argument.

$$F_T(t) = P(T \leq t) = P(T \leq 4) = \int_0^{0.9} 1 - \left((1/0.9)e^{-(1-y)t} \right) dy \text{ for } t > 0.$$

Then:

$$f_T(t) = \frac{dF_T(t)}{dt} = (1/0.9) \left(\frac{(0.1)e^{-0.1t}}{t} + \frac{e^{-0.1t}}{t^2} - \frac{e^{-t}}{t} - \frac{e^{-t}}{t^2} \text{ for } t > 0. \right)$$

and $f_T(t) = 0$ otherwise.

4. What is the probability that the network congestion level was over 1/2 given that the download time is over 2 minutes ($P(Y > 1/2|T > 2)$)?

$$P(Y > 1/2|T > 2) = \frac{P(T > 2|Y > 1/2)P(Y > 1/2)}{P(T > 2)}.$$

5. What is the probability density function of the network congestion level Y given that the download time is t minutes?

$$f_{Y|T}(y) = \frac{f_{T|Y}(t)f_Y(y)}{f_T(t)}.$$

Problem 4.

A small business has daily profits that are random and independent. Assume that the daily profit X is a continuous random variable between 500 and 1500 TL. The business is open for sale for 200 days in a year. Note that $E[X] = 1000$ TL and $Var(X) = 83,333.33$ ($\sqrt{Var(X)} = 288.675$).

1. Use the normal approximation to compute the probability that the yearly profit is over 220,000 TL.

Let Π be the profit. Note that

$$\Pi = \sum_{i=1}^{200} X_i$$

Because Π is the sum of a large number of i.i.d. random variables, Π must have a normal distribution.

$$E[\Pi] = 200E[X_i] = 200,000 \text{ TL}$$

and

$$Var(\Pi) = 200Var[X_i] = 200(Var(X)) = 1.67 \times 10^7 \text{ TL}$$

So Π is normal with $\mu = 200000$ and $\sigma = \sqrt{Var(\Pi)} = 4082.48$.

$$P(\Pi > 220000) = 1 - \phi\left(\frac{220000 - 200000}{4082.48}\right) = 0.$$

2. The owner realizes that profits are above average on Fridays. Let the daily profit on Fridays be a continuous uniform random variable Y between 1300 and 1500. Note that $E[Y] = 1400$ TL and $Var(Y) = 3,333.33$ ($\sqrt{Var(Y)} = 57.735$). Assume that there are 160 regular days (with profits X) and 40 Fridays (with profits Y) in a year. Use the normal approximation to compute the probability that the yearly profit is over 220,000 TL under this assumption.

Note that

$$E[\Pi] = (160)1000 + (40)1400 = 216000$$

$$Var(\Pi) = (160)83333.33 + (40)3333.33 = 1.34 \times 10^7$$

This time Π is normal with $\mu = 216000$ and $\sigma = \sqrt{Var(\Pi)} = 3669.7$.

$$P(\Pi > 220000) = 1 - \phi\left(\frac{220000 - 216000}{3669.7}\right) = 0.138.$$

Problem 5.

Short questions (provide full explanations of your answers).

1. The Sariyer Storm and the Kilyos Thunder are playing the regional basketball play-off finals. Each team has an equal probability of winning any game (ties are not allowed) and the first team to reach 3 wins, wins the series. Assume that Sariyer has won the first two games. What is the probability that Sariyer wins the series (i.e. reaches three wins before Kilyos)?

Sariyer wins in the following ways: win game 3, lose game 3 and win game 4 and lose game 4 and win game 5. The total probability is then: $1/2 + 1/4 + 1/8 = 7/8$.

2. There are two coins (blue and red). The blue coin is unbiased and the red one is biased and has $P(\text{Heads}) = 3/4$. You select a coin randomly and give it your friend without looking at its color. Your friend tosses it and tells you that the outcome is Heads. What is the probability that you chose the red coin?

$$P(R|H) = \frac{P(H|R)P(R)}{P(H)} = \frac{(3/4)(1/2)}{(3/4)(1/2) + (1/2)(1/2)} = \frac{3}{5}.$$

3. Assume that there are four exams in a course and you think (before taking the exams) that your grade in each of the exams is a normal random variable with mean 80 and standard deviation 10. What is the probability that you receive a grade higher than 90 in at least three of the four exams (assuming that the grades in different exams are independent)?

$$P(X_i > 90) = P\left(Z > \left(\frac{90 - 80}{10}\right)\right) = 1 - \Phi(1) = 0.159.$$

Then, let A be the event that at least three of the four exams have grades higher than 90. We obtain

$$P(A) = 4(0.159)^3(0.841) + (0.159)^4.$$

Problem 6.

Alice and Bob take turns tossing a biased coin that turns up heads with probability p . If Alice tosses a head, she wins the game. If Bob tosses a tail, he wins the game. The coin tossings continue until one of the two players wins. Alice tosses first.

Useful formula: For any real number a such that $|a| < 1$, we have

$$1 + a + a^2 + a^3 + \dots = \frac{1}{1 - a}.$$

- (a) Find the probability that Alice wins the game. (Hint: the answer is not p)

Solution:

The scenarios from Alice to win the game are

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THH

THTHH

THTHTHH

.....

Thus, we get

$$\mathbb{P}[\text{Alice wins}] = p [1 + p(1 - p) + p^2(1 - p)^2 + p^3(1 - p)^3 + \dots] = \frac{p}{1 - p(1 - p)}.$$

- (b) Now, find the probability that Bob wins the game from first principles, that is, do not just write down $\mathbb{P}(\text{Bob wins}) = 1 - \mathbb{P}(\text{Alice wins}) = 1 - \{\text{answer from part (a)}\}$.

Solution:

For Bob, he wins if

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THTHTHTT

Thus, we get

$$\mathbb{P}[\text{Bob wins}] = (1-p)^2 [1 + p(1-p) + p^2(1-p)^2 + \dots] = \frac{(1-p)^2}{1-p(1-p)}.$$

- (c) Show that your answers to parts (a) and (b) add up to 1.

Solution:

$$\frac{p}{1-p(1-p)} + \frac{(1-p)^2}{1-p(1-p)} = \frac{p+1-2p+p^2}{1-p(1-p)} = 1.$$

- (d) For what value(s) of p is this a fair game (that is, a game that both Alice and Bob have equal probability $1/2$ of winning)?

Solution:

We know $\frac{p}{1-p+p^2} = \frac{1}{2}$. It implies $p^2 - p + 1 = 0$, which yields $p = \frac{3-\sqrt{5}}{2}$.

- (e) Now consider once again the general game (with arbitrary value of p), and let X denote the total number of coin tosses in the game. Thus, X is a discrete random variable taking on values 1, 2, 3, ... Find the probability mass function $P_X(m)$ of the random variable X .

Solution:

An odd X implies Alice winning, while an even X implies Bob winning.

$$P_X(m) = \begin{cases} [p(1-p)]^{\frac{m-1}{2}} p, & m = 1, 3, 5, \dots \\ [p(1-p)]^{\frac{m-2}{2}} (1-p)^2, & m = 2, 4, 6, \dots \end{cases}$$

- (f) Find $\mathbb{E}[X]$ as a function of p .

Solution:

$$\begin{aligned} \mathbb{E}X &= \sum_{m=1}^{\infty} m P_X(m) \\ &= \sum_{k=0}^{\infty} (2k+1) [p(1-p)]^k p + \sum_{k=0}^{\infty} (2k+2) [p(1-p)]^k (1-p)^2 \\ &= \frac{p + 2(1-p)^2}{1-p(1-p)} + [2p + 2(1-p)^2] + \sum_{k=0}^{\infty} k [p(1-p)]^k \\ &= \frac{p + 2(1-p)^2}{1-p(1-p)} + [2p + 2(1-p)^2] \frac{p(1-p)}{[1-p(1-p)]^2}. \end{aligned}$$

Problem 7.

Check the appropriate box for each of the statements below. No justification is required.

- (a) Answer “TRUE” if the statement below is true for ALL events A and B with $0 < \mathbb{P}(A) < 1$ and $0 < \mathbb{P}(B) < 1$. Otherwise answer “FALSE”.

TRUE FALSE

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☐ | $\mathbb{P}(A \cup B) \geq \max\{\mathbb{P}(A), \mathbb{P}(B)\}$ |
| ☐ | ☒ | $\mathbb{P}(A \cap B) \geq \min\{\mathbb{P}(A), \mathbb{P}(B)\}$ |
| ☐ | ☒ | $\mathbb{P}(A B)\mathbb{P}(B) + \mathbb{P}(A^c B)\mathbb{P}(B) = \mathbb{P}(A)$ |
| ☐ | ☒ | If $\mathbb{P}(A B) = \mathbb{P}(B A)$, then $\mathbb{P}(A) = \mathbb{P}(B)$ |
| ☐ | ☒ | If $A \cap B = \emptyset$, then A and B are independent. |
| ☒ | ☐ | If $\mathbb{P}(A B) = \mathbb{P}(A)$, then $\mathbb{P}(B^c A) = 1 - \mathbb{P}(B)$ |

- (b) X is a continuous random variables with finite mean and variance. The functions $F_X(x)$ and $f_X(x)$ denote the cumulative distribution function and the probability density function of X , respectively. Answer “TRUE” if the statement is true for all such random variables X , otherwise answer “FALSE”.

TRUE FALSE

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|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | $f_X(x) \leq 1$ for all x . |
| ☒ | ☐ | There is some real number a such that $2F_X(a) > F_X(2a)$. |
| ☒ | ☐ | $\mathbb{P}(a \leq X < b) = \mathbb{P}(e^a \leq e^X < e^b)$ for all real numbers $a < b$. |
| ☒ | ☐ | $\mathbb{P}(X^2 - 2X + 1 < -1) = 0$. |
| ☒ | ☐ | If $Y = X $, then $\mathbb{E}[Y] \geq \mathbb{E}[X]$. |
| ☒ | ☐ | If $Y = X $, then $\text{var}[Y] \leq \text{var}(X)$. |

- (c) X and Y are random variables such that $\text{var}(X) = \text{var}(Y) = \sigma^2 < \infty$. In addition, we have that $\text{var}(2X + 3Y + 4) = \text{var}(3X - 2Y + 1)$. Answer “TRUE” if the statement is true for all such random variables X, Y , otherwise answer “FALSE”.

TRUE FALSE

- ☐ ☒ $\mathbb{E}[X^2] = \mathbb{E}[Y^2]$
- ☒ ☐ X and Y are uncorrelated random variables.
- ☐ ☒ X and Y are independent random variables.
- ☒ ☐ $\text{var}(2X + 3Y + 4) = \text{var}(2X - 3Y + 1)$
- ☒ ☐ $\text{cov}(X + Y, X - Y) = 0$
- ☒ ☐ $\text{var}(X + Y) = 2\sigma^2$

- (d) **[12 points]** Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables. Let Y_n and Z_n be two random variables defined as

$$Y_n = X_1 + X_2 + \dots + X_n,$$

$$Z_n = |X_1| + |X_2| + \dots + |X_n|,$$

respectively. Assume that the random variables $X_1, X_2, \dots, X_n$ have finite mean μ and finite variance σ^2 , and that $\mathbb{E}[|X_1|] = \tau$.

TRUE FALSE

- ☒ ☐ $\text{var}(|X_n|) = \sigma^2 + \mu^2 - \tau^2$.
- ☐ ☒ For sufficiently large n , Y_n and Z_n are approximately jointly Gaussian.
- ☒ ☐ For sufficiently large n , we have $\mathbb{P}(Y_n - n\mu \leq Z_n - n\tau) \approx 0.5$.
- ☐ ☒ If $X_1, X_2, \dots, X_n$ are Gaussian, then $\mathbb{P}(Y \leq n) = \mathbb{P}(X_1 \leq 1)$
- ☒ ☐ $\mathbb{P}(Z_n \geq n) \leq \tau$ for all n .

Problem 8.

Check the appropriate box for each of the statements below. No justification is required.

- (a) Answer “TRUE” if the statement below is true for ALL events A and B with $0 < \mathbb{P}(A), \mathbb{P}(B) < 1$. Otherwise answer “FALSE”.

TRUE FALSE

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☐ | $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$ |
| ☒ | ☐ | $\mathbb{P}(A B) \leq 1$ |
| ☐ | ☒ | $\mathbb{P}(A B) + \mathbb{P}(A B^c) = 1$ |
| ☐ | ☒ | $\frac{\mathbb{P}(A B)}{\mathbb{P}(B)} \leq 1$ |
| ☒ | ☐ | $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$ |
| ☒ | ☐ | $\mathbb{P}(A B) = \mathbb{P}(B A)\mathbb{P}(A)/\mathbb{P}(B)$ |

- (b) Answer “TRUE” if the statement is true for all continuous random variables X with finite mean and variance, otherwise answer “FALSE”.

TRUE FALSE

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|-------------------------------------|-------------------------------------|---|
| ☐ | ☒ | If $\text{var}(X) = 0$ then $\mathbb{E}[X] = 0$. |
| ☒ | ☐ | $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$ |
| ☐ | ☒ | $g(X)$ is a continuous random variable for all functions $g(\cdot)$. |
| ☒ | ☐ | If $Y = X $, then $f_Y(v) = f_X(v) + f_X(-v)$ for all $v > 0$. |

- (c) Let X and Y be *independent* Gaussian random variables with parameters $\mathbb{E}[X] = \mu_X$, $\mathbb{E}[Y] = \mu_Y$ and $\text{var}(X) = \sigma_X^2$, $\text{var}(Y) = \sigma_Y^2$. Indicate whether each of the following statements is true or false.

Notation: $f_{XY}(x, y)$, $f_X(x)$, $f_Y(y)$ denote the joint and marginal PDFs of X and Y , respectively; $\Phi(x)$ is the CDF of a standard Gaussian random variable with zero mean and unit variance, that is,

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.$$

Finally, $\text{cov}(X, Y)$ is the covariance of the two random variables.

TRUE FALSE

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☐ | $f_{XY}(x, y) = f_X(x)f_Y(y)$ |
| ☐ | ☒ | $\mathbb{E}[XY] = 0$ |
| ☒ | ☐ | $\text{var}(X + Y) = \sigma_X^2 + \sigma_Y^2$ |
| ☒ | ☐ | $\mathbb{P}(X + Y < z) = \Phi\left(\frac{z - \mu_X - \mu_Y}{\sqrt{\sigma_X^2 + \sigma_Y^2}}\right)$ |
| ☐ | ☒ | $\mathbb{P}(X \leq x Y \leq y) = \Phi(x)$ |
| ☒ | ☐ | $\text{cov}(X, Y) = 0$ |

- (d) Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (IID) random variables. Let Y_n be a random variable defined as

$$Y_n = X_1 + X_2 + \dots + X_n.$$

Assume that the random variables $X_1, X_2, \dots, X_n$ have finite mean μ and finite variance σ^2 .

TRUE FALSE

- | | | |
|-------------------------------------|-------------------------------------|---|
| ☒ | ☐ | If $\hat{X}_n = Y_n/n$, then $\text{var}(\hat{X}_n) = \sigma^2/n$. |
| ☐ | ☒ | Y_n is a Gaussian random variable. |
| ☒ | ☐ | For sufficiently large n , we have $\mathbb{P}(Y_n \leq n\mu + \sigma\sqrt{n}) \approx \Phi(1)$. |
| ☒ | ☐ | If $\mu = \sigma = 1$, then the variance of the product $X_1 \cdot X_2 \cdots X_n$ is $2^n - 1$. |

Problem 9.

The random point (X, Y) is uniformly distributed on the shaded region shown below.

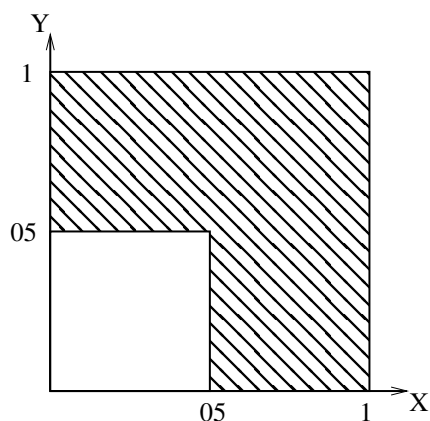
- (a) The joint PDF of X, Y can be written as

$$f_{XY}(x, y) = \begin{cases} c, & \text{if } (x, y) \text{ is in the shaded region;} \\ 0, & \text{otherwise.} \end{cases}$$

What is the value of the constant c ?

Solution:

$$c = \frac{1}{4} = \frac{4}{3}$$



- (b) Find the marginal PDF $f_X(x)$ of the random variable X . In order to obtain full credit, you must specify the value of $f_X(x)$ for all real numbers x .

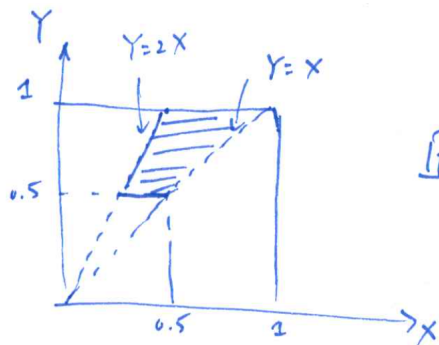
Solution:

$$f_X(x) = \begin{cases} \frac{2}{3}, & \text{if } 0 < x < 0.5, \\ \frac{4}{3}, & \text{if } 0.5 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (c) Find $\mathbb{P}(X < Y < 2X)$.

Solution:

$$\mathbb{P}[X < Y < 2X] = \frac{4}{3} \times \text{Area in the figure below} = \frac{4}{3} \cdot \frac{\frac{1}{2} + \frac{1}{4}}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

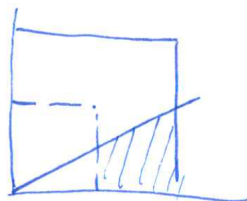


- (d) Find the PDF of the random variable $Z = Y/X$. In order to obtain full credit, you must specify the value of $f_Z(z)$ for all real numbers z .

Solution:

We know that

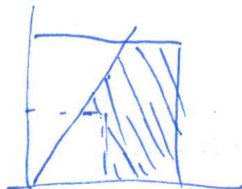
$$F_Z(z) = \mathbb{P}\left(\frac{Y}{X} \leq z\right) = \mathbb{P}(Y \leq zX).$$



If $z \leq 1$ (see the figure above), we get $F_Z(z) = \frac{4}{3} \cdot \frac{\frac{z}{2} + z}{2} \cdot \frac{1}{2} = \frac{z}{2}$.

If $z > 1$ (see the figure below), we get $F_Z(z) = 1 - \frac{4}{3} \cdot \frac{\frac{1}{2z} + \frac{1}{z}}{2} \cdot \frac{1}{2} = 1 - \frac{1}{2z}$, which yields

$$f_Z(z) = \begin{cases} \frac{1}{2}, & 0 < z < 1 \\ \frac{1}{2}z^{-2}, & z \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$



Problem 10.

On any given flight, an airline's goal is to fill the plane as much as possible without overbooking. If, on average, 10% of customers cancel their tickets, all independently of each other, what is the probability that a particular flight will be overbooked if the airline sells 320 tickets for a plane that has a maximum capacity of 300 people? What is the probability that a plane with maximum capacity 150 people will be overbooked if the airline sells 160 tickets? Express the answers in terms of the standard normal cumulative distribution function, $\Phi(x)$. (*Hint.* Consider the probability p of a success to be the probability that a given person actually shows up for the flight. From this, deduce what the effective mean and standard deviation are for the number of successes in the case when $n = 320$ or when $n = 160$, respectively.)

Solution:

Let N be the number of people that actually show up for the flight. The flight will be overbooked if $N > 300$. We can use the normal approximation to the binomial to calculate this probability. Note that $n = 320$ and $p = 0.9$ where p is the probability of a success (a customer does show up for the flight). So the effective mean is $0.9 \cdot 320 = 288$ and the effective standard deviation is $\sqrt{320 \cdot 0.9 \cdot 0.1} = 5.367$. Then we have,

$$P(301 \leq N \leq 320) = \Phi\left(\frac{320 - 288}{5.367}\right) - \Phi\left(\frac{301 - 288}{5.367}\right) = 1 - \Phi(2.4181) = 0.0078.$$

A similar computation for the smaller plane shows that the effective mean and standard deviation are 144 and 3.794, respectively, and that the probability of overbooking in that case is:

$$P(151 \leq N \leq 160) = 1 - \Phi\left(\frac{151 - 144}{3.794}\right) = 1 - \Phi(1.8450) = 0.0329.$$

Problem 11.

Two coins, one fair and one coming up heads with probability $1/3$, are in a pocket. Other than this difference in their bias, the coins are indistinguishable. One coin is drawn from the pocket, with each of the two coins equally likely to be chosen. We want to know which coin we actually picked by flipping it multiple times and observing the results.

- (a) Suppose the coin drawn is flipped three times. What is the probability that heads show up on all three flips?

Solution:

$$\mathbb{P}[\text{HHH}] = \frac{1}{2} \cdot \left(\frac{1}{2}\right)^3 + \frac{1}{2} \left(\frac{1}{3}\right)^3.$$

- (b) Given that a total of two heads show up on the three flips, what is the probability that the coin is fair?

Solution:

$$\mathbb{P}[\text{fair} | 2H] = \frac{\mathbb{P}[\text{fair}] \mathbb{P}[2H | \text{fair}]}{\mathbb{P}[2H]} = \frac{\frac{1}{2} \binom{3}{2} \left(\frac{1}{2}\right)^3}{\frac{1}{2} \binom{3}{2} \left(\frac{1}{2}\right)^3 + \frac{1}{2} \binom{3}{2} \left(\frac{1}{3}\right)^2 \frac{2}{3}}$$

(No need to simplify the answer.)

- (c) If you flip the selected coin 1000 times, what is the probability that the number of heads is no greater than 250? You just need to get an approximate answer in terms of the function $\phi(x)$, defined as

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.$$

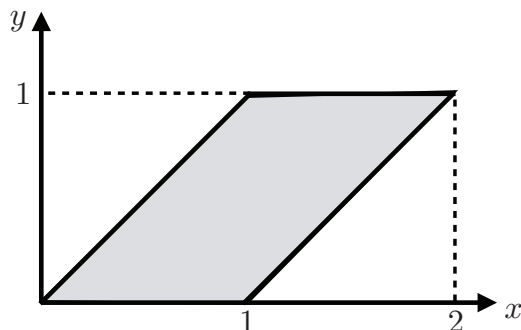
Solution:

$$\begin{aligned} & \mathbb{P}[X_1 + \dots + X_{1000} \leq 250] \\ &= \frac{1}{2} \mathbb{P}[X_1 + \dots + X_{1000} \leq 250 | \text{fair}] + \frac{1}{2} \mathbb{P}[X_1 + \dots + X_{1000} \leq 250 | \text{biased}] \\ &= \frac{1}{2} \mathbb{P}\left[\frac{X_1 + \dots + X_{1000} - 500}{\sqrt{1000 \cdot \frac{1}{4}}} \leq \frac{250 - 500}{\sqrt{1000 \cdot \frac{1}{4}}}\right] \\ &\quad + \frac{1}{2} \mathbb{P}\left[\frac{X_1 + \dots + X_{1000} - 1000/3}{\sqrt{1000 \cdot \frac{1}{3} \cdot \frac{2}{3}}} \leq \frac{250 - 1000/3}{\sqrt{1000 \cdot \frac{1}{3} \cdot \frac{2}{3}}}\right] \\ &= \frac{1}{2} \phi(-\sqrt{250}) + \frac{1}{2} \phi\left(-\frac{5}{2}\sqrt{5}\right) \end{aligned}$$

Problem 12.

Let X and Y be two continuous random variables with joint pdf

$$f_{X,Y}(x,y) = \begin{cases} c, & \text{if } (x,y) \text{ is in the shaded parallelogram region} \\ 0, & \text{otherwise.} \end{cases}$$



- (a) What is the value of the constant c ?

Solution:

$$c = \frac{1}{\text{Area}} = 1.$$

- (b) Are X and Y independent? Justify your answer.

Solution:

No. For example, if $X = 1$, Y can take values from $[0, 1]$ and if $X = 1.5$, Y can only take values from $[0.5, 1]$.

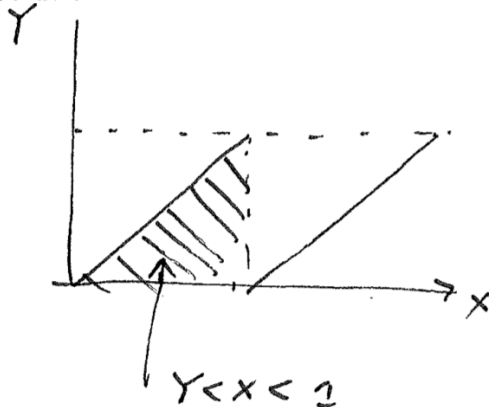
- (c) Find the marginal pdf $f_X(x)$. To obtain full credit, you must specify the value of $f_X(x)$ for all x , $-\infty < x < \infty$.

Solution:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dy = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

- (d) Find $\mathbb{P}(Y < X < 1)$.

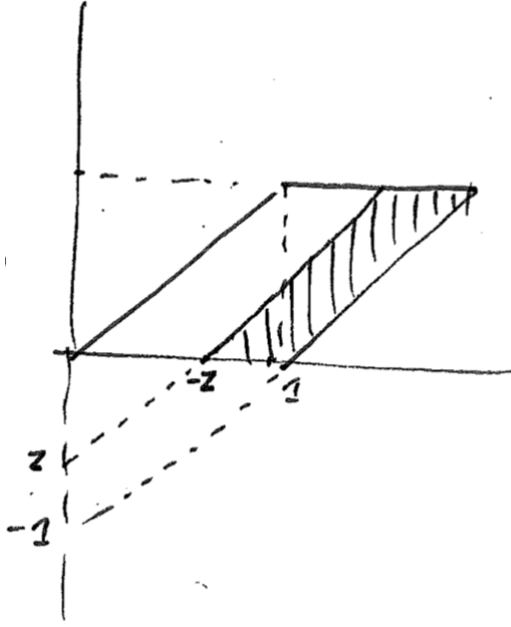
Solution:



From the figure above, we get $\mathbb{P}[Y < X < 1] = \frac{1}{2}$.

- (e) Find the CDF of the random variable $Z = Y - X$. In order to obtain full credit, you must specify the value of $F_Z(z)$ for all real numbers z .

Solution:



From the figure above, we get

$$F_Z(z) = \mathbb{P}(Y - X \leq z) = \mathbb{P}(Y \leq X + z) = \begin{cases} 0, & z < -1 \\ 1, & z > 0 \\ 1 + z, & -1 < z < 0 \end{cases}$$